Introduction to Probability Theory (Math 2502) Spring 2026
Shagnik Das

Exercise Sheet 6

B13902024 Chang, Yi-Kuei
Due date: 13:00, May 26th.
Submissions that are late will receive no credit.

You should try to solve all of the exercises below — each problem is worth 10 points. Make sure to clearly justify your answers, defining everything precisely and stating any results you are using. You should then submit your solutions in Gradescope on NTU COOL, either as a typed PDF or a clear scan of your handwritten answers, and should indicate where in the file the solutions to each exercise are. If you have difficulties with Gradescope, please send your solutions by e-mail.

Exercise 1 There is a chess match consisting of 12 games between Grandmaster Ju Wenjun and Grandmaster Vaishali R. Ju Wenjun wins the first game and loses the second. Thereafter, she is very much a confidence-based player, in the sense that her probability of winning a game is proportional to the number of games she has won so far. More precisely, for $k \geq 2$,

$$\mathbb{P}\left(\text{Ju Wenjun wins game } k+1 \mid \text{Ju Wenjun has won } x \text{ of the first } k \text{ games }\right) = \frac{x}{k}.$$

Let X be the random variable counting the number of games Ju Wenjun will win in the 12-game match. Determine $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution: Let X_k denote the number of games Ju Wenjun has won after k games have been played. We are given the initial conditions:

$$X_1 = 1, \quad X_2 = 1$$

We want to find the expected value $\mathbb{E}[X_{12}]$ and the variance $\mathrm{Var}(X_{12})$.

For $k \geq 2$, the conditional probabilities describing the transition from game k to game $k+1$ are given by:

$$\mathbb{P}(X_{k+1} = x+1 \mid X_k = x) = \frac{x}{k}$$

$$\mathbb{P}(X_{k+1} = x \mid X_k = x) = 1 - \frac{x}{k} = \frac{k-x}{k}$$

1. Distribution of X_k

Claim: For any $k \geq 2$, X_k is uniformly distributed over the integers $\{1, 2, \ldots, k-1\}$. That is,

$$\mathbb{P}(X_k = j) = \frac{1}{k-1} \quad \text{for } j = 1, 2, \ldots, k-1$$

Proof. We prove the claim by mathematical induction on k.

Base Case: For $k = 2$, since $X_2 = 1$ deterministically, we have:

$$\mathbb{P}(X_2 = 1) = 1 = \frac{1}{2-1}$$

Thus, the base case holds.

Inductive Step: Assume that the claim holds for some $k \geq 2$. That is,

$$\mathbb{P}(X_k = j) = \frac{1}{k-1} \quad \text{for } j = 1, 2, \ldots, k-1$$

We want to show that for $k+1$:

$$\mathbb{P}(X_{k+1} = j) = \frac{1}{k} \quad \text{for } j = 1, 2, \ldots, k$$

We analyze this by partitioning into three cases based on the value of j:

Case 1: $2 \leq j \leq k-1$. By the law of total probability, the event $X_{k+1} = j$ can occur either if Ju Wenjun had j wins and lost the $(k+1)$-th game, or if she had $j-1$ wins and won the $(k+1)$-th game:

$$\begin{aligned}
\mathbb{P}(X_{k+1} = j) &= \mathbb{P}(X_k = j)\mathbb{P}(X_{k+1} = j \mid X_k = j) + \mathbb{P}(X_k = j-1)\mathbb{P}(X_{k+1} = j \mid X_k = j-1) \\
&= \left(\frac{1}{k-1} \cdot \frac{k-j}{k}\right) + \left(\frac{1}{k-1} \cdot \frac{j-1}{k}\right) \\
&= \frac{1}{k(k-1)} \left((k-j) + (j-1)\right) \\
&= \frac{k-1}{k(k-1)} = \frac{1}{k}
\end{aligned}$$

Case 2: $j = k$. Ju Wenjun can only reach k wins after $k+1$ games if she already had $k-1$ wins after k games and wins the $(k+1)$-th game:

$$\begin{aligned}
\mathbb{P}(X_{k+1} = k) &= \mathbb{P}(X_k = k-1)\mathbb{P}(X_{k+1} = k \mid X_k = k-1) \\
&= \frac{1}{k-1} \cdot \frac{k-1}{k} = \frac{1}{k}
\end{aligned}$$

Case 3: $j = 1$. Ju Wenjun can only have exactly 1 win after $k+1$ games if she had 1 win after k games and loses the $(k+1)$-th game:

$$\begin{aligned}\mathbb{P}(X_{k+1} = 1) &= \mathbb{P}(X_k = 1)\mathbb{P}(X_{k+1} = 1 \mid X_k = 1) \\ &= \frac{1}{k-1} \cdot \frac{k-1}{k} = \frac{1}{k}\end{aligned}$$

Combining all three cases, we have shown that $\mathbb{P}(X_{k+1} = j) = \frac{1}{k}$ for all $j \in \{1, 2, \ldots, k\}$. This completes the inductive step. □

2. Computation of Expectation and Variance for $k = 12$

Applying the proven claim to the 12-game match ($k = 12$), X_{12} is uniformly distributed over $\{1, 2, \ldots, 11\}$:

$$\mathbb{P}(X_{12} = j) = \frac{1}{11} \quad \text{for } j = 1, 2, \ldots, 11$$

Expected Value: Using the identity for the sum of the first n integers $\sum_{j=1}^{n} j = \frac{n(n+1)}{2}$:

$$\mathbb{E}[X_{12}] = \sum_{j=1}^{11} j \cdot \mathbb{P}(X_{12} = j) = \frac{1}{11}\sum_{j=1}^{11} j = \frac{1}{11} \cdot \frac{11 \cdot 12}{2} = 6$$

Variance: Using the formula $\mathrm{Var}(X_{12}) = \mathbb{E}[X_{12}^2] - (\mathbb{E}[X_{12}])^2$, and the identity for the sum of squares $\sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}$:

$$\begin{aligned}\mathrm{Var}(X_{12}) &= \left(\sum_{j=1}^{11} j^2 \cdot \frac{1}{11}\right) - 6^2 \\ &= \frac{1}{11} \cdot \left(\frac{11 \cdot 12 \cdot 23}{6}\right) - 36 \\ &= \frac{11 \cdot 12 \cdot 23}{11 \cdot 6} - 36 \\ &= 2 \cdot 23 - 36 \\ &= 46 - 36 = 10\end{aligned}$$

Conclusion

The expected number of wins in the 12-game match is **6**, and the variance is **10**.

Exercise 2 Two friends are meant to meet at the library to study together for their probability exam. Their arrival times are independent and uniformly distributed between 6pm and 7pm. Let W denote the amount of time the first friend to arrive has to wait for the second friend. Determine $\mathbb{E}[W]$ and $\mathbb{P}(W \geq 15)$.

Solution: Let F_1 and F_2 denote the arrival times of the two friends, measured in minutes past 6:00 pm. Based on the problem statement, F_1 and F_2 are independent and identically distributed (*i.i.d.*) continuous random variables uniformly distributed over the interval $[0, 60]$. Thus, we have:

$$F_1, F_2 \overset{\text{i.i.d.}}{\sim} \text{Unif}[0, 60]$$

The marginal probability density functions (PDFs) are given by:

$$f_{F_1}(x) = \frac{1}{60}, \quad x \in [0, 60] \quad \text{and} \quad f_{F_2}(y) = \frac{1}{60}, \quad y \in [0, 60]$$

By independence, their joint probability density function $f_{F_1,F_2}(x, y)$ is:

$$f_{F_1,F_2}(x, y) = f_{F_1}(x) \cdot f_{F_2}(y) = \frac{1}{3600}, \quad (x, y) \in [0, 60] \times [0, 60]$$

Let W denote the waiting time of the first friend to arrive for the second friend. This can be mathematically expressed as the absolute difference between their arrival times:

$$W = |F_1 - F_2|, \quad \text{where } 0 \leq W \leq 60$$

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1. Calculating the Expected Waiting Time $\mathbb{E}[W]$

By the law of the unconscious statistician (LOTUS), the expected value of W is:

$$\mathbb{E}[W] = \int_0^{60} \int_0^{60} |x - y| \, f_{F_1,F_2}(x, y) \, dx \, dy$$

Substituting the joint PDF yields:

$$\mathbb{E}[W] = \int_0^{60} \int_0^{60} |x - y| \cdot \frac{1}{3600} \, dx \, dy$$

To evaluate this, we split the inner integral to remove the absolute value sign based on the regions $x \geq y$ and $x < y$:

$$\mathbb{E}[W] = \frac{1}{3600} \int_0^{60} \left(\int_y^{60} (x - y) \, dx + \int_0^{y} (y - x) \, dx \right) dy$$

Evaluating the inner integrals with respect to x:

$$\int_y^{60} (x - y) \, dx = \left[\frac{1}{2}x^2 - xy \right]_y^{60} = (1800 - 60y) - \left(\frac{1}{2}y^2 - y^2 \right) = 1800 - 60y + \frac{1}{2}y^2$$
$$\int_0^{y} (y - x) \, dx = \left[yx - \frac{1}{2}x^2 \right] 0^y = \left(y^2 - \frac{1}{2}y^2 \right) - 0 = \frac{1}{2}y^2$$

Summing the two inner integral results:

$$\left(1800 - 60y + \frac{1}{2}y^2\right) + \frac{1}{2}y^2 = y^2 - 60y + 1800$$

Now, substituting this back into the outer integral with respect to y:

$$\begin{aligned}
\mathbb{E}[W] &= \frac{1}{3600}\int_0^{60} \left(y^2 - 60y + 1800\right) dy \\
&= \frac{1}{3600}\left[\frac{1}{3}y^3 - 30y^2 + 1800y\right]_0^{60} \\
&= \frac{1}{3600}\left(\frac{1}{3}(60)^3 - 30(60)^2 + 1800(60)\right) \\
&= \frac{1}{3600}\left(\frac{1}{3}\cdot 216000 - 30\cdot 3600 + 108000\right) \\
&= \frac{1}{3600}(72000 - 108000 + 108000) \\
&= \frac{72000}{3600} = 20
\end{aligned}$$

Thus, the expected waiting time is **20 minutes**.

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2. Calculating the Probability $\mathbb{P}(W \geq 15)$

We want to find the probability that the waiting time is at least 15 minutes:

$$\mathbb{P}(W \geq 15) = \mathbb{P}(|F_1 - F_2| \geq 15)$$

This event can be decomposed into two disjoint scenarios: either friend 1 arrives at least 15 minutes after friend 2, or friend 2 arrives at least 15 minutes after friend 1.

$$\mathbb{P}(W \geq 15) = \mathbb{P}(\{F_1 \geq F_2 + 15\} \cup \{F_2 \geq F_1 + 15\})$$

By the axioms of probability for disjoint events and by symmetry (since F_1, F_2 are i.i.d.):

$$\mathbb{P}(W \geq 15) = \mathbb{P}(F_1 \geq F_2 + 15) + \mathbb{P}(F_2 \geq F_1 + 15) = 2\,\mathbb{P}(F_1 \geq F_2 + 15)$$

We express this probability as a double integral over the region where $x \geq y + 15$ within the $[0, 60] \times [0, 60]$ square. The bounds for y range from 0 to 45 (since $x \leq 60 \implies y + 15 \leq 60 \implies y \leq 45$), and the bounds for x range from $y + 15$ to 60:

$$\mathbb{P}(W \geq 15) = 2\int_0^{45}\int_{y+15}^{60} f_{F_1,F_2}(x, y)\, dx\, dy$$

Substituting the joint PDF:

$$\begin{aligned}
\mathbb{P}(W \geq 15) &= 2 \int_0^{45} \int_{y+15}^{60} \frac{1}{3600}\, dx\, dy \\
&= \frac{1}{1800} \int_0^{45} \Big[x\Big]_{y+15}^{60} dy \\
&= \frac{1}{1800} \int_0^{45} (60 - (y+15))\, dy \\
&= \frac{1}{1800} \int_0^{45} (45 - y)\, dy
\end{aligned}$$

Evaluating the final definite integral:

$$\begin{aligned}
\mathbb{P}(W \geq 15) &= \frac{1}{1800} \left[45y - \frac{1}{2}y^2\right]_0^{45} \\
&= \frac{1}{1800} \left(45(45) - \frac{1}{2}(45)^2\right) \\
&= \frac{1}{1800} \left(\frac{1}{2}(45)^2\right) \\
&= \frac{2025}{3600} = \frac{9}{16}
\end{aligned}$$

Thus, the probability that the first friend to arrive waits at least 15 minutes is $\mathbf{\frac{9}{16}}$ (or **0.5625**).

Exercise 3 The number of daily visitors to a popular chess website follows a Poisson distribution with rate λ. One of the ways this website makes a profit is by showing ads to its users. Each visitor clicks on an ad with probability p, independently of all other visitors. Letting X denote the number of times an ad is clicked on in a day, determine $\mathbb{P}(X = k)$ for all $k \geq 0$, and deduce the value of $\mathbb{E}[X]$.

Solution: Suppose P is the number of visitors to the website, then $P \sim \text{Poi}(\lambda)$. For any $k \geq 0$, then by total law of probability and countable additivity:

$$
\begin{aligned}
\mathbb{P}(X=k) &= \sum_{s=0}^{\infty} \mathbb{P}(P=s, X=k) = \sum_{s=0}^{\infty} \mathbb{P}(X=k \mid P=s)\mathbb{P}(P=s) \\
&= \sum_{s=k}^{\infty} \binom{s}{k} p^k (1-p)^{s-k} \cdot \frac{e^{-\lambda}\lambda^s}{s!} = \sum_{s=k}^{\infty} \frac{s!}{k!(s-k)!} p^k (1-p)^{s-k} \cdot \frac{e^{-\lambda}\lambda^s}{s!} \\
&= \frac{p^k e^{-\lambda}}{k!} \sum_{s=k}^{\infty} \frac{(1-p)^{s-k}\lambda^s}{(s-k)!} = \frac{p^k e^{-\lambda}}{k!} \sum_{m=0}^{\infty} \frac{(1-p)^m \lambda^{m+k}}{m!} \\
&= \frac{\lambda^k p^k e^{-\lambda}}{k!} \sum_{m=0}^{\infty} \frac{\big((1-p)\lambda\big)^m}{m!} = \frac{(\lambda p)^k e^{-\lambda}}{k!} \cdot e^{(1-p)\lambda} = \frac{(\lambda p)^k e^{-\lambda+\lambda-p\lambda}}{k!} = \frac{(\lambda p)^k e^{-p\lambda}}{k!}
\end{aligned}
$$

This shows $X \sim \mathrm{Poi}(\lambda p)$. Hence,

$$\mathbb{E}[X] = \lambda p.$$

Exercise 4 On Saturday, three siblings, Alpha, Beta, and Omega, will run up the 101. Their running times, measured in minutes, are independent; Alpha's is exponentially distributed with rate $\frac{1}{27}$, Beta's is exponentially distributed with rate $\frac{1}{33}$, and Omega's is uniformly distributed over $(25, 35)$.

Let X denote the time the fastest sibling takes, Y the time the second-fastest sibling takes, and Z the time the slowest sibling takes. Determine $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[Z]$.

Solution: Let A, B, O be the running times of Alpha, Beta, and Omega, respectively. Thus,

$$A \sim \mathrm{Exp}\left(\frac{1}{27}\right), \qquad B \sim \mathrm{Exp}\left(\frac{1}{33}\right), \qquad O \sim \mathrm{Unif}(25, 35),$$

and assume that A, B, O are independent. Let

$$X = \min(A, B, O), \qquad Y = \text{the middle value of } \{A, B, O\}, \qquad Z = \max(A, B, O).$$

Write

$$\lambda_A = \frac{1}{27}, \qquad \lambda_B = \frac{1}{33}, \qquad \lambda = \lambda_A + \lambda_B = \frac{20}{297}.$$

We will compute $\mathbb{E}[X]$, $\mathbb{E}[Y]$, and $\mathbb{E}[Z]$.

Note that for a non-negative random variable T, we have

$$\mathbb{E}[T] = \int_0^{\infty} \mathbb{P}(T \geq t)\, dt - \int_0^{\infty} \mathbb{P}(T \leq -t)\, dt = \int_0^{\infty} \mathbb{P}(T \geq t)\, dt.$$

Step 1: The expectation of X.

Since $X = \min(A, B, O)$ and the variables are independent,

$$\mathbb{P}(X > t) = \mathbb{P}(A > t)\mathbb{P}(B > t)\mathbb{P}(O > t).$$

Now

$$\mathbb{P}(A > t) = e^{-\lambda_A t}, \qquad \mathbb{P}(B > t) = e^{-\lambda_B t},$$

and for $O \sim \mathrm{Unif}(25, 35)$,

$$\mathbb{P}(O > t) = \begin{cases} 1, & 0 \le t < 25, \\ \dfrac{35 - t}{10}, & 25 \le t \le 35, \\ 0, & t > 35. \end{cases}$$

Therefore

$$\mathbb{P}(X > t) = \begin{cases} e^{-\lambda t}, & 0 \le t < 25, \\ e^{-\lambda t}\dfrac{35 - t}{10}, & 25 \le t \le 35, \\ 0, & t > 35. \end{cases}$$

Using $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t)\, dt$, we get

$$\mathbb{E}[X] = \int_0^{25} e^{-\lambda t}\, dt + \frac{1}{10}\int_{25}^{35} (35 - t)e^{-\lambda t}\, dt.$$

A direct computation gives

$$\mathbb{E}[X] = \frac{1 - e^{-25\lambda}}{\lambda} + \frac{e^{-25\lambda}(10\lambda - 1) + e^{-35\lambda}}{10\lambda^2}.$$

Substituting $\lambda = \frac{20}{297}$,

$$\mathbb{E}[X] = \frac{297}{20} + \frac{88209}{4000}\left(e^{-700/297} - e^{-500/297}\right).$$

Step 2: The expectation of Z.

Since $Z = \max(A, B, O)$,

$$\mathbb{E}[Z] = \int_0^\infty \mathbb{P}(Z > t)\, dt.$$

We determine $\mathbb{P}(Z > t)$ piecewise.

Note that

$$\mathbb{P}(Z \le t) = \mathbb{P}(\max\{A, B, O\} \le t) = \mathbb{P}(A \le t, B \le t, O \le t) = \mathbb{P}(A \le t)\mathbb{P}(B \le t)\mathbb{P}(O \le t)$$

by independence.

If $0 \le t < 25$, then $O > t$ is always true, so $Z > t$ always. Hence

$$\mathbb{P}(Z > t) = 1.$$

If $25 \le t \le 35$, then

$$\mathbb{P}(Z \le t) = \mathbb{P}(A \le t)\mathbb{P}(B \le t)\mathbb{P}(O \le t) = (1 - e^{-\lambda_A t})(1 - e^{-\lambda_B t})\frac{t - 25}{10},$$

so

$$\mathbb{P}(Z > t) = 1 - (1 - e^{-\lambda_A t})(1 - e^{-\lambda_B t})\frac{t - 25}{10}.$$

If $t > 35$, then $O \le t$ is always true, hence

$$\begin{aligned}
\mathbb{P}(Z > t) &= 1 - \mathbb{P}(Z \le t) = 1 - \mathbb{P}(A \le t, B \le t, O \le t) \\
&= 1 - \mathbb{P}(A \le t)\mathbb{P}(B \le t)\mathbb{P}(O \le t) \\
&= 1 - (1 - e^{-\lambda_A t}) \cdot (1 - e^{-\lambda_B t}) \cdot 1 = e^{-\lambda_A t} + e^{-\lambda_B t} - e^{-\lambda t}.
\end{aligned}$$

Therefore

$$\begin{aligned}
\mathbb{E}[Z] = \int_0^{25} 1\, dt + \int_{25}^{35} \left[1 - (1 - e^{-\lambda_A t})(1 - e^{-\lambda_B t})\frac{t - 25}{10}\right] dt \\
+ \int_{35}^{\infty} \left(e^{-\lambda_A t} + e^{-\lambda_B t} - e^{-\lambda t}\right) dt.
\end{aligned}$$

Now for any $r > 0$,

$$\int_{25}^{35} \frac{t - 25}{10} e^{-rt}\, dt + \int_{35}^{\infty} e^{-rt}\, dt = \frac{e^{-25r} - e^{-35r}}{10r^2}.$$

Using this with $r = \lambda_A, \lambda_B, \lambda$, we obtain

$$\mathbb{E}[Z] = 30 + \frac{e^{-25\lambda_A} - e^{-35\lambda_A}}{10\lambda_A^2} + \frac{e^{-25\lambda_B} - e^{-35\lambda_B}}{10\lambda_B^2} - \frac{e^{-25\lambda} - e^{-35\lambda}}{10\lambda^2}.$$

Substituting $\lambda_A = \frac{1}{27}$, $\lambda_B = \frac{1}{33}$, and $\lambda = \frac{20}{297}$ gives

$$\mathbb{E}[Z] = 30 + \frac{729}{10}\left(e^{-25/27} - e^{-35/27}\right) + \frac{1089}{10}\left(e^{-25/33} - e^{-35/33}\right) - \frac{88209}{4000}\left(e^{-500/297} - e^{-700/297}\right).$$

Step 3: The expectation of Y.

Since X, Y, Z are the order statistics of A, B, O,

$$X + Y + Z = A + B + O.$$

Taking expectations and using linearity,

$$\mathbb{E}[X] + \mathbb{E}[Y] + \mathbb{E}[Z] = \mathbb{E}[A] + \mathbb{E}[B] + \mathbb{E}[O].$$

Now

$$\mathbb{E}[A] = 27, \qquad \mathbb{E}[B] = 33, \qquad \mathbb{E}[O] = \int_{25}^{35} t \cdot \frac{1}{10}\, \mathrm{d}t = 30,$$

so

$$\mathbb{E}[X] + \mathbb{E}[Y] + \mathbb{E}[Z] = 90.$$

Hence

$$\mathbb{E}[Y] = 90 - \mathbb{E}[X] - \mathbb{E}[Z].$$

Substituting the formulas found above,

$$\mathbb{E}[Y] = \frac{903}{20} - \frac{729}{10}\left(e^{-25/27} - e^{-35/27}\right) - \frac{1089}{10}\left(e^{-25/33} - e^{-35/33}\right) + \frac{88209}{2000}\left(e^{-500/297} - e^{-700/297}\right).$$

Final answers.

$$\boxed{\mathbb{E}[X] = \frac{297}{20} + \frac{88209}{4000}\left(e^{-700/297} - e^{-500/297}\right)}$$

$$\boxed{\mathbb{E}[Y] = \frac{903}{20} - \frac{729}{10}\left(e^{-25/27} - e^{-35/27}\right) - \frac{1089}{10}\left(e^{-25/33} - e^{-35/33}\right) + \frac{88209}{2000}\left(e^{-500/297} - e^{-700/297}\right)}$$

$$\boxed{\mathbb{E}[Z] = 30 + \frac{729}{10}\left(e^{-25/27} - e^{-35/27}\right) + \frac{1089}{10}\left(e^{-25/33} - e^{-35/33}\right) - \frac{88209}{4000}\left(e^{-500/297} - e^{-700/297}\right)}$$